# Meric Altug Gemalmaz, Ph.D.

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## EDUCATION

**Purdue University, West Lafayette, Indiana**

- Ph.D. in Computer Science (Human-AI Interaction Focused) **3.96/4.0** — Jan. 2020 – June 2025
- M.S. in Computer Science (Transferred to Ph.D.) **3.93/4.0** — Jan. 2019 – Dec. 2019
- B.S. in Computer Science (Software Engineering Track, **Distinction**) **3.92/4.0** — Aug. 2015 – Dec. 2018

## SKILLS

**Programming Languages:** Python, C/C++, C#, JavaScript, Java, HTML/CSS, R, SQL, Bash
**Machine Learning:** Large-scale retrieval and ranking, recommender systems, content-integrity and quality signals, LLM-based signal design and prompting, model evaluation, A/B experimentation and metric design
**Machine Learning Tools:** scikit-learn, GSL, NumPy, Pandas
**Tools & Platforms:** Git, Meteor.js, MongoDB, MTurk, Android SDK, Firebase, GDB

## EXPERIENCE

**ML Research Scientist, Meta, Menlo Park, CA** — Aug. 2025 – Present

- Work on Facebook Groups' large-scale ranking systems: content recommendation and content quality.
- In roughly nine months on Groups, shipped **17** A/B-tested launches across the group recommendation and unconnected group post recommendation stacks, building LLM-based content-quality signals, **two** new candidate generators, and content-integrity detection signals.
- Learned the group and unconnected post recommendation stacks in my first weeks, took production oncall by month **four**; chased root causes in retrieval and measurement instead of patching symptoms.
- In the Meta AI Org, create and improve training data for AI models, build the workflows that produce it, and evaluate AI agent systems.

**Web App Development for Data Collection, Purdue** — Jan. 2020 – May 2025

- Coordinated **1,500+** human participants across **4+** research projects on **MTurk** for data collection and analysis.
- Developed and deployed web applications using **JavaScript** and **Meteor.js**, managed back-end databases with **MongoDB**, and implemented front-end interfaces using **HTML** and **CSS**.
- Delivered **90%** participant satisfaction and blocked **80%** of the bot attacks the system received by designing scenario-based, interactive UIs.

**Fairness in AI-Driven Gig Work: User-Centric Insights, Purdue** — Dec. 2023 – May 2025

- Initiated a collaborative study with gig workers on the long-term impact of AI-driven gig assignments at varying fairness levels, yielding actionable insights for fairer algorithmic job management.

**Design Implications of Fairness in Human-AI Interactions, Purdue [P1, P3]** — Mar. 2021 – Dec. 2023

- Conducted **six** human-subject experiments on how loan applicants' repeated interactions with an AI-based loan approval system affect their fairness perceptions and willingness to keep engaging with the AI.
- Simulated **1,000** loan applicant interactions using a **Markov Decision Process** in **Python** to estimate experiment parameters; collected real-world data and applied **regression analysis** in **R** to uncover key human-AI interaction trends.
- Found crucial fairness insights: over **75%** of users continue using biased models out of necessity rather than perceived fairness, urging developers to rethink usage metrics as indicators of model fairness.

**Data Bias Mitigation Algorithm, Purdue [P2]** — Mar. 2020 – Mar. 2021

- Leveraged unsupervised learning techniques to detect and mitigate cognitive bias in crowdsourced data annotations.
- Applied **probabilistic graphical models** and **Expectation-Maximization** with the **GSL** library in **C** to reduce annotator bias, improving label accuracy across **100** simulated and **12** real-world tasks.
- Achieved over **10%** increase in inferred label accuracy over existing baselines through reduction in annotation bias.

## HONORS AND AWARDS

Recipient of the Graduate Teaching Award, Purdue — Nov. 2022
Recipient of the Raymond Boyce Graduate Teaching Award, Purdue — Apr. 2020
Dean's List and Semester Honors (**8 Semesters**), Purdue — Aug. 2015 – Dec. 2018

## SELECTED PAPERS

**[P1] Meric Altug Gemalmaz**, Ming Yin. "Understanding Decision Subjects' Fairness Perceptions and Retention in Repeated Interactions with AI-Based Decision Systems." *Proceedings of the 5th AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, Oxford, UK, Aug. 2022.

**[P2] Meric Altug Gemalmaz**, Ming Yin. "Accounting for Confirmation Bias in Crowdsourced Label Aggregation." *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*, Online, Aug. 2021.

**[P3] Meric Altug Gemalmaz**, Ming Yin. "Understanding Decision Subjects' Engagement with and Perceived Fairness of AI Models When Opportunities of Qualification Improvement Exist." *arXiv:2410.03126*